JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY GURAJADA VIZIANAGARAM

SRI SIVANI COLLEGE OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

★ LAB MANUAL WITH TEST CASES & FLOWCHARTS ★

Programming for Problem Solving using C (PPS) Lab Manual & Code Repository

Course Code:	R23-ES-LAB1	**Academic Scheme:**	I B.Tech I Semester (R23 Regulation)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Week 1: Familiarization with Linux / GCC & Basic I/O

Writing, compiling and executing simple C programs, Standard I/O, Escape sequences, Data types and format specifiers.

Week 2: Conditional Statements & Branching

Implementation of if-else ladders, switch-case, nested conditions: Quadratic equation roots, Grade calculation, Simple calculator.

Week 3: Loops & Iterative Formulations

While, do-while, for loops: Prime number checks, Fibonacci sequences, Armstrong numbers, Pattern printing exercises.

Week 4: Arrays & Matrix Operations

1D arrays: Linear and binary search, Bubble sort, 2D arrays: Matrix addition, multiplication, and transpose calculation.

Week 5: Pointers, Functions & File Management

Pass-by-value vs pass-by-reference, Recursion (Tower of Hanoi), Dynamic memory allocation (malloc, calloc, free), File read/write operations.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Continuous Lab Assessment	15 Marks
Model Exam	15 Marks
External Viva & Execution	50 Marks

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM